Unit 1: Polynomial and Rational Functions

Name: ___________________________ Date: ________________ Period: _________

Learning Targets — I Can...

1. I can rewrite a polynomial or rational expression in factored or standard form and explain what each form reveals about the function (zeros, end behavior, asymptotes, holes).
2. I can perform polynomial long division to express $f(x) = g(x)q(x) + r(x)$, identify a slant asymptote from the quotient, and use Pascal's Triangle to expand $(x + c)^n$.

What's in This Packet

#	Component	Description	Score
1	Warm-Up	Spiral review: factoring, rational zeros, asymptotes, and holes	_______
2	Guided Notes	Factored vs. standard form; polynomial long division; binomial theorem	_______
3	Group Activity	Form-switching with rational functions (tiered R / A / E)	_______
4	Exit Ticket	Choose a form; execute one long-division step; Pascal expansion	_______
5	Homework	Mixed: forms, long division, slant asymptotes, binomial theorem	_______
		Total	_______